JUNIOR SCHOOL LEARNING HUB

GRADE 9 INTEGRATED SCIENCE

CONTINUOUS ASSESSMENT TEST 1 (CAT 1)

TERM 1

Time: 1 hour 15 minutes **Total: 50 marks**

Learner's Name: ..... Grade: 9 Date: .....

INSTRUCTIONS

1. Answer ALL questions.
2. Show calculations and give units where necessary.
3. Read each question carefully before answering.
4. Use diagrams where they help to explain your answer.

SECTION A: SHORT-ANSWER QUESTIONS (20 MARKS)

1. Define an atom. (2 marks)

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2. Name the three main subatomic particles and state the charge of each. (3 marks)

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3. State the relative location of protons, neutrons and electrons in an atom. (3 marks)

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4. An atom has atomic number 11 and mass number 23. Determine the number of protons, electrons and neutrons in a neutral atom. (3 marks)

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5. What is an isotope? (2 marks)

.....

6. State two physical properties commonly associated with metals. (2 marks)

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7. Name one metal that is liquid at room temperature. (1 mark)

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8. What is an alloy? (2 marks)

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9. Give one reason why alloys are often preferred to pure metals. (1 mark)

.....

10. State one use of stainless steel and identify one metal present in it. (1 mark)

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SECTION B: STRUCTURED QUESTIONS (30 MARKS)

11. Structure of the atom (8 marks)

(a) Draw a simple labelled model of an atom showing the nucleus and electrons. (3 marks)

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(b) Explain the meaning of atomic number and mass number. (2 marks)

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(c) An atom contains 17 protons, 18 neutrons and 17 electrons. Determine its atomic number and mass number. (2 marks)

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(d) Explain why an atom is electrically neutral when it has equal numbers of protons and electrons. (1 mark)

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12. Electron arrangement and isotopes (6 marks)

(a) Write the electron arrangement of a neutral atom with atomic number 12 using shells. (2 marks)

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(b) Explain why atoms of the same element may have different mass numbers. (2 marks)

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(c) State one practical application of isotopes. (2 marks)

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13. Properties and uses of metals (6 marks)

(a) Explain the terms malleability and ductility. (2 marks)

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(b) Give two reasons why copper is used in electrical wiring. (2 marks)

.....

(c) State two ways of preventing or reducing corrosion of iron. (2 marks)

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14. Metals and alloys (5 marks)

(a) Distinguish between a pure metal and an alloy. (2 marks)

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(b) Name the main metals present in brass and bronze. (2 marks)

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(c) State one use of brass or bronze. (1 mark)

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15. Selecting materials for a purpose (5 marks)

A school wants to make a strong, corrosion-resistant outdoor handrail.

(a) Suggest a suitable metal or alloy for the handrail. (1 mark)

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(b) Give two reasons for your choice. (2 marks)

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(c) Explain why ordinary uncoated iron may be unsuitable for this purpose. (2 marks)

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END OF QUESTION PAPER

JUNIOR SCHOOL LEARNING HUB

GRADE 9 INTEGRATED SCIENCE – CAT 1

MARKING SCHEME | TERM 1

Total: 50 marks. Award marks for scientifically correct equivalent responses. Accept clear labelled diagrams.

SECTION A: SHORT-ANSWER QUESTIONS (20 MARKS)

1. An atom is the smallest unit of an element that retains the chemical identity/properties of that element. (2 marks)

2. Proton: positive (+); neutron: neutral (0); electron: negative (−). (1 mark each)

3. Protons and neutrons are in the nucleus; electrons occupy shells/energy levels around the nucleus. (3 marks)

4. Protons = 11; electrons = 11; neutrons = 23 − 11 = 12. (1 mark each)

5. Atoms of the same element with the same atomic number but different numbers of neutrons/mass numbers. (2 marks)

6. Any two: lustrous/shiny, malleable, ductile, good conductors of heat, good conductors of electricity, generally strong. (1 mark each)

7. Mercury. (1 mark)

8. A mixture of a metal with one or more other elements, at least one of which is a metal; accept suitable school-level definition. (2 marks)

9. Any one: greater strength/hardness, improved corrosion resistance, improved durability or another valid property. (1 mark)

10. Accept cutlery, kitchen equipment, sinks or surgical instruments; chromium and iron are present (nickel may also be present). Award use ½ and metal ½. (1 mark)

SECTION B: STRUCTURED QUESTIONS (30 MARKS)

11. Structure of the atom (8 marks)

- (a) Nucleus at centre; protons and neutrons identified in nucleus; electrons shown around nucleus/on shells. Award 1 mark for each correct feature, maximum 3.
- (b) Atomic number = number of protons; mass number = total number of protons and neutrons. (1 mark each)
- (c) Atomic number = 17; mass number = 17 + 18 = 35. (1 mark each)
- (d) Protons have positive charge and electrons have equal negative charge; equal numbers cancel to give zero net charge. (1 mark)

12. Electron arrangement and isotopes (6 marks)

- (a) 2,8,2. Award 2 marks for the correct arrangement; 1 mark for a correct attempt with one error.
- (b) They have the same number of protons but different numbers of neutrons. (2 marks)
- (c) Any one valid application with explanation: cobalt-60 in cancer radiotherapy; iodine-131 in diagnosis/treatment of thyroid conditions; carbon-14 in dating once-living materials; suitable industrial tracing applications. (2 marks)

13. Properties and uses of metals (6 marks)

- (a) Malleability: ability to be hammered/rolled into sheets; ductility: ability to be drawn into wires. (1 mark each)
- (b) Copper is a good electrical conductor and is ductile/flexible for drawing into wires. (1 mark each)
- (c) Any two: painting, oiling/greasing, galvanising, electroplating, plastic coating or keeping dry. (1 mark each)

14. Metals and alloys (5 marks)

- (a) Pure metal consists of one metallic element; alloy is a mixture of a metal with one or more other elements. (1 mark each)
- (b) Brass: copper and zinc; bronze: copper and tin. (1 mark each)
- (c) Accept musical instruments, medals, ornaments, taps/fittings, bearings or other valid use. (1 mark)

15. Selecting materials for a purpose (5 marks)

- (a) Stainless steel or another appropriate corrosion-resistant alloy/material. (1 mark)
- (b) Any two: resists corrosion, strong, durable, requires less maintenance, suitable for outdoor exposure. (1 mark each)
- (c) Iron reacts with oxygen and moisture to form rust; rust weakens the handrail and continued exposure causes deterioration. (2 marks)

Teacher's administration and scope note

- This CAT assesses introductory content on atomic structure, isotopes, metals, alloys and selection of materials.
- Before administering, check the school's Term 1 scheme and confirm that all assessed content has been taught.
- If the class has not yet covered isotopes or alloys, remove or replace those items and adjust the mark allocation.
- Use practical demonstrations only with appropriate supervision and safety precautions. Do not allow learners to handle hazardous chemicals or hot metals without approved procedures.

Prepared by Junior School Learning Hub. Original teacher-prepared assessment; not an official KICD examination.